

ANGELA KANG

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EDUCATION

University of Southern California (USC)

Los Angeles, CA

M.S. Applied Data Science | GPA: 4.0

August 2025 – May 2027(expected)

• **Relevant Courses:** Machine Learning for Data Science, Foundations of Data Management

University of California, Santa Barbara (UCSB)

Goleta, CA

B.S. Statistics and Data Science

September 2021 – June 2025

• **Relevant Courses:** Probability & Statistics, Machine Learning, Multivariate Analysis, Time Series, Stochastic Processes, Linear Algebra, Real Analysis, Numerical Methods, Data Structures & Algorithms, Experimental Design, Python Programming

EXPERIENCE & LEADERSHIP

Hackathon Winner, SBHacks XI | JavaScript, HTML/CSS, Chrome Extension APIs

• Led front-end development of My Meowcho, a Chrome extension that gamifies productivity with animated cats, built with a 4-person team; won Best Overall Hack out of 60+ teams.

PROJECTS

APS Failure Prediction | Python, scikit-learn, XGBoost, SMOTE

- Predicted truck air-pressure-system failures from 170 sensors, where only 1.67% of 60K records were failures and a missed failure costs 50× a false alarm, so accuracy alone was misleading.
- Compared Random Forest and XGBoost with class weights and SMOTE, keeping oversampling and imputation inside the training folds to avoid data leakage.
- Best model (XGBoost + SMOTE) cut missed failures 38% (106 → 66) vs. baseline Random Forest, raised recall to 82%, and lowered the dataset's cost metric by 37%.

Jute Pest Classification | Python, TensorFlow/Keras, Transfer Learning

- Automated identification of 17 jute pest species from ~6,800 images, a task that normally needs field experts.
- Benchmarked 5 ImageNet-pretrained CNNs with the same classification head, data augmentation, early stopping, and mixed-precision GPU training; DenseNet201 reached 96.6% test accuracy (F1 0.97) vs. 83.1% for the next best.

Learning Paradigms: Supervised to Active Learning | Python, scikit-learn, SVM, Clustering

- Tested how many labels a tumor classifier really needs (569 tumors, 30 features) by comparing L1-SVM, semi-supervised self-training, k-means, and spectral clustering over 30 Monte Carlo runs.
- Self-training with half the labels reached 96.2% test accuracy vs. 96.7% fully supervised; on banknote data, active learning (labeling points nearest the SVM margin) beat random sampling with fewer labels.

COVID Custom SQL Engine | Python (no pandas), Streamlit, Plotly

- Built core database operations from scratch (CSV parser, filter, projection, group-by, hash join) on a custom column-based table to learn how SQL engines work.
- Ran it on a 350K-row (92 MB) COVID dataset joined with country data and deployed a live Streamlit dashboard of cases, deaths, and vaccination trends.

Reddit Topic Clustering & Embedding Comparison | Python, BeautifulSoup, MySQL, gensim, scikit-learn

- With a 3-person team, built a pipeline that collects posts from 4 tech-career subreddits, cleans the text, stores it in MySQL, and refreshes on a schedule.
- Ran the Doc2Vec experiments (3 vector sizes) and compared them with Word2Vec “bag-of-bins” vectors on ~5,000 posts using KMeans; Word2Vec bins clustered best (silhouette 0.31 vs. 0.13).

Paper Airplane Flight Experiment | R, Experimental Design, ANOVA

- Designed and ran a randomized complete block experiment (30 airplanes × 4 paperclip placements), then used a power analysis to confirm 30 blocks gave 80% power.
- ANOVA showed placement mattered ($F = 31.8$, $p < 0.001$); Tukey HSD found a middle clip flew 27.8 in. farther than the control, while a rear clip flew 25.6 in. shorter.

SKILLS & INTERESTS

Languages: Python, R, SQL, C, CUDA, JavaScript, HTML, MATLAB, LaTeX, C++, React

Data Science & Analytics: Predictive modeling, statistical modeling, regression analysis, time series analysis, experimental design, ANOVA, machine learning, deep learning, transfer learning, exploratory data analysis (EDA), feature engineering, dimensionality reduction (PCA), imbalanced-data methods (SMOTE), clustering, relational data modeling, custom SQL-style operations (filter, select, groupby, join)

Tools & Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Git/GitHub, Jupyter, Excel, Streamlit, MySQL, MongoDB, Apache Spark, Hadoop, cuBLAS, BeautifulSoup, Selenium, Tesseract OCR, GCC, NVCC

Technical Skills: GPU programming & parallel computing, CUDA kernel development, high-performance computing, Linux/Ubuntu development, web scraping, data pipeline development, database querying, SQL joins and aggregations, NoSQL-style data handling, distributed data processing concepts, Python bindings to C/CUDA code, statistical analysis, data visualization